

Comparing British and American Audio Description of Movies

Igor Sterner, Alex Lascarides, Frank Keller



THE UNIVERSITY *of* EDINBURGH
informatics

Narrative in Movies

Here is a scene in *Oppenheimer* (2023), with a narration of key visual events



[https://www.dropbox.com/scl/fi/21b51xl6v7qxs2q606s5d/
apple.mp4?rlkey=h5elszdo83ipcx8azo45onds2&st=qyeadydw&dl=0](https://www.dropbox.com/scl/fi/21b51xl6v7qxs2q606s5d/apple.mp4?rlkey=h5elszdo83ipcx8azo45onds2&st=qyeadydw&dl=0)

Audio Description (AD) as a narrative art form

- AD is a verbal narration of key visual information necessary to understand a video
- It provides access to blind and visually impaired audiences
- It tells the story the director decided to tell

Audio description translates a visual narrative into a verbal one

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Audio description translates a visual narrative into a verbal one

- AD is a narrative and literary art form
- The degree of creative license is hotly debated
- Regional guidelines settle parts of this debate

- Ultimately, all perception is interpretive and complete objectivity is impossible
- AD makes the underlying processes of narrative construction in movies explicit, and comparing AD versions makes the subjectivity in these processes explicit
- We aim to understand and model movie narratives (and ultimately generate AD)

Comparing British and American Audio Description of Movies

- Prior comparisons of British and American AD relied on theory and anecdote
- We provide the first **quantitative** comparison of the two regions

Audio description guidelines – a comparison

Hansjörg Bittner, University of Hildesheim

Introduction

In 2007, Gert Vercauteren published his comparison of audio description (AD) guidelines, *Towards a European Guideline for Audio Description*, in which he suggests the creation of one European AD standard. Three years later, the Royal National Institute of Blind People came up with *A Comparative Study of Audio Description Guidelines Prevalent in Different Countries* (Royal National Institute of Blind People 2010), a document that provides a good overview of the similarities and differences in AD standards. The present paper compares guidelines from Australia, France, Germany, Ireland, the United Kingdom, and the United States. The aim is to critically analyse them and pave the way for a set of rules which, if followed, would produce a universally “good” audio description (AD). As the various guidelines from the above countries do not adhere to a uniform structure, we need to align them first with the help of an overall theoretical framework, before we can match them against each other.



1. We define hypotheses from the **regional guidelines** and from an **AD practitioner**
2. We design automatic tests and apply them to a controlled corpus

UK and US descriptions of the Oppenheimer scene

In the bunker, Oppenheimer removes his goggles, blinking rapidly and squinting to see the explosion growing into a column of fire that blooms high into the dark sky, taking on a mushroom shape. [US]

UK and US descriptions of the Oppenheimer scene

In the bunker, Oppenheimer removes his goggles, blinking rapidly and squinting to see the explosion growing into a column of fire that blooms high into the dark sky, taking on a mushroom shape. [US]

Oppenheimer takes off his goggles. An immense fireball billows into the night sky. Oppenheimer's eyelids flicker open. The explosion climbs higher and higher, white heat at its core darkening at the edges into an orange mushroom shape. [UK]

UK and US descriptions of the Oppenheimer scene

In the bunker, Oppenheimer removes his goggles, blinking rapidly and squinting to

- Both versions describe the same events, largely objectively
- The language is almost poetic and paints a vivid scene
- There is some flexibility in when events are described, and which events are selected for description

heat at its core darkening at the edges into an orange mushroom shape.

[UK]

Hypothesis: scene changes and lexical preferences

Scene changes

Inside a tent, Lawrence drinks from a bottle and passes it to Frank. [UK]

After nightfall, warm lantern [...]. Lawrence passes a liquor bottle [...] [US]

In a tent at night, warm light [...]. Lawrence passes a bottle of booze [...]

[US→UK post-edit]

Hypothesis: scene changes and lexical preferences

Lexical preferences

Inside a tent, Lawrence drinks from a **bottle** and passes it to Frank. [UK]

After nightfall, warm lantern [...]. Lawrence passes a **liquor bottle** [...] [US]

In a tent at night, warm light [...]. Lawrence passes a **bottle** of booze [...]

[US→UK post-edit]

Hypothesis: aspect and voice

Aspect

Oppenheimer **is standing** at the lectern. [UK]

Oppenheimer **stands** at a lectern. [US]

At the front, Oppenheimer **stands** at a lectern looking slightly nervous. [US→UK]

post-edit]

Voice

Oppenheimer sees a torch **being shone** at cars parked on a suburban street. [UK]

Oppenheimer notices someone with a flashlight checking out a car [...]. [US]

Oppenheimer notices someone shining a torch by a car and making notes. [US→UK]

post-edit]

Hypothesis: subjectivity

Subjectivity

Oppenheimer is standing at the lectern. [UK]

Oppenheimer stands at a lectern. [US]

At the front, Oppenheimer stands at a lectern **looking slightly nervous**. [US→UK post-edit]

Naming in advance

A smile from **Evans**, the oldest board member. [UK]

An **elderly member of the board** smiles.

[+8 mins] Robb shakes hands with the chairman, the elderly **Evans** and [...]. [US]

An older member of the board, **Ward Evans**, smiles. [US→UK post-edit]

Hypothesis: Dialogue Overlap

Example

As classmates leave a lab, one shouts to Oppenheimer, *“Don’t forget to clean up!”*

- **US AD** voices over this dialogue entirely to describe a green apple (note the apple soon becomes salient to the narrative).
- **UK AD** does not overlap

The REFRAME dataset

(note that this dataset is presented in another paper, which is currently in submission)

- Professional American AD, British AD and dialogue subtitles for 206 movies
- (also: 3K+ scenes from the movies, aligned screenplays, and SDH)
- Fully public release for scene-based data; academic release for full-movie data

	British AD	American AD
Total token count	1,137,598	1,419,576
Total sentence count	121,123	171,224
Average sentences per movie	588	831
Average tokens per movie	5,522	6,891

The REFRAME dataset

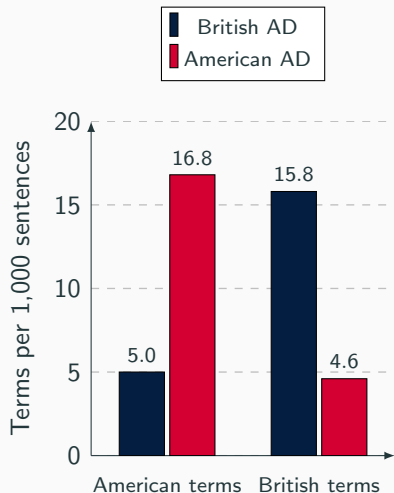
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- Parse every AD and dialogue sentence with Stanza: (tokenization, sentence segmentation, universal dependencies and POS tags, OntoNotes named entities)
- Match against pre-defined lists of American and British terms
- One automatic test per hypothesis, e.g. present progressive via a present-tense auxiliary with a participle; passive voice via passive dependency relations
- All tests are relatively simple; see paper for more details

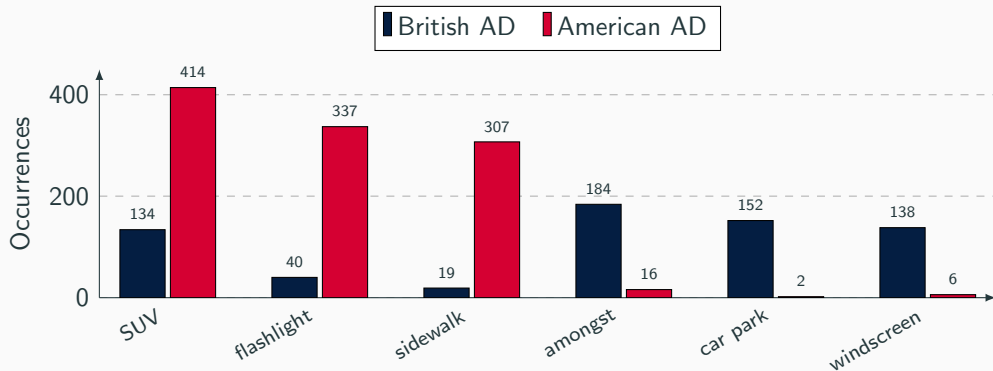
Results: Lexicon



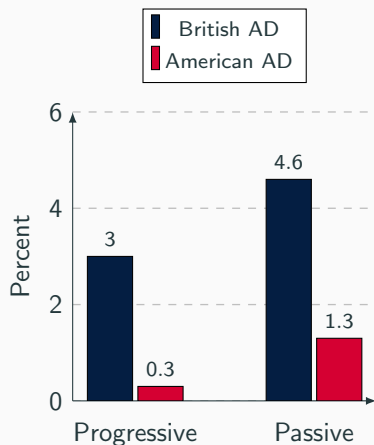
(all differences in upcoming results significant)

- Each region uses its own terms about three times more frequently
- Differences only relevant for c. 1% of sentences

Results: Lexical Examples

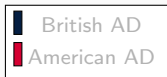


Results: Aspect and Voice



- British AD uses the progressive aspect about ten times more often
- Passive voice is also more frequent in British AD
- US guidelines are more explicit than UK in using *simple* language

Results: Subjectivity



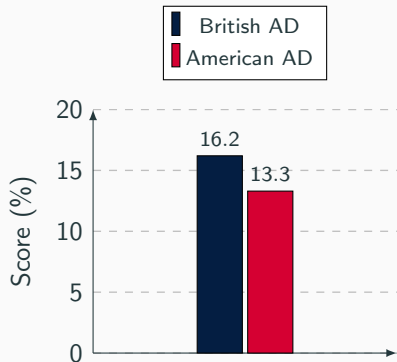
- British AD has higher average adjective subjectivity

- Score adjectives using a lexicon of subjectivity ratings
- Adverbial modifiers also contribute to the sentence score
- Average sentence-level scores within each movie



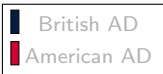
this word, negatively stating that it is vague and evaluative evaluative.

Results: Subjectivity



- British AD has higher average adjective subjectivity.
- *beautiful* occurs almost three times as often in British AD
- US guidelines directly address the use of this word, negatively stating that it is vague and evaluative.

Results: Subjectivity



- British AD has higher average adjective

he turns back to find a **beautiful** woman glaring at him. [UK]

then looks back and finds a **dark-haired young** woman standing before him. [US]

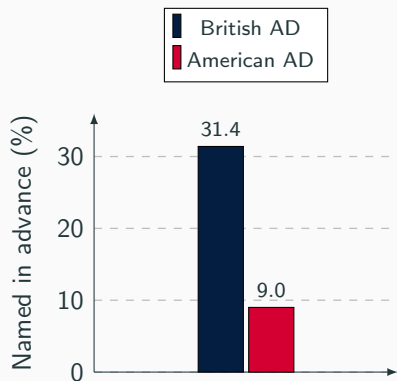
Bridget notices the church is mainly filled with **beautiful** young women. [UK]

Bridget notices many of the pews filled with **fashionable** young women. [US]



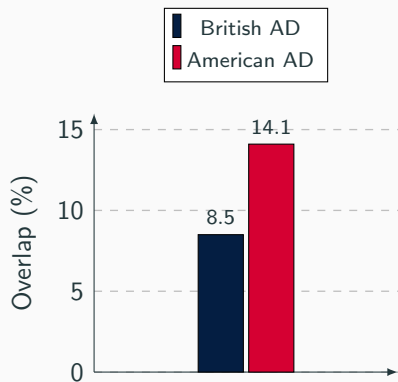
vague and evaluative evaluative.

Results: Naming in Advance



- British AD names characters in advance more frequently
- In British AD, the average named-in-advance character is 11 mins earlier, vs. 7 mins in US AD
- Our analysis shows that it still occurs

Results: Dialogue Overlap



- American AD overlaps subtitles and music lyrics more often
- Then again, it is also longer overall

Results: Cueing Scene Changes

We train a model to detect scene boundaries from AD text alone, then test on each region.

Recall	UK	US
Trained on UK	55.4	72.4
Trained on US	42.9	66.0
Trained on UK+US	55.1	69.0

- Recall is significantly lower on UK AD
- The model learns explicit cues that are more frequent in American AD
- British AD verbalises scene changes less explicitly

Take-Home Messages

- British and American AD show small but statistically distinguishable differences in lexicon, aspect, voice, subjectivity, naming, dialogue overlap, and scene cueing
- Every difference is motivated by the regional guidelines, except lexical variation, which had no prior theoretical treatment
- AD is a tightly constrained narrative technique that makes narrative construction in movies explicit, an attractive domain for narratology
- This work is part of my PhD on AD generation